Greedy Packing of Nested Rings: Superadditivity, Placement Rules, and a Tribonacci Threshold

[Author names]*

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Abstract

Motivated by the humble act of frying squid rings, we study the packing of annuli ("rings") with common width w into a disk, where a ring may be nested inside the hole of a strictly larger ring; this is a circular-container, selection-oriented variant of the Recursive Circle Packing Problem arising in tube logistics. Two natural objectives – the number of placed rings and their total contact area – genuinely diverge, and we locate the divergence exactly: contact area is a *superadditive* function of the radius, cardinality is not. Our main results concern the descending greedy algorithm. If the radii are *superincreasing* (each exceeds the sum of all smaller ones), then (i) the selection greedy maximizes $\sum v(r_i)$ for *every* monotone superadditive value v, and (ii) the placement rule is irrelevant: any greedy that puts each ring into an arbitrary feasible container attains the lexicographically maximal feasible set. Both statements hold for containers of arbitrary shape and in every dimension. We then prove sharpness: placement irrelevance holds unconditionally for $n \leq 3$ rings but fails at $n = 4$, via an explicit counterexample with a rigorous non-packability proof; moreover, two *twin instances* sharing an identical prefix show that no deterministic placement rule that is a function of the observable state succeeds on all instances, and every randomized rule fails some instance with probability at least 1/2. Measuring the violation of superincreasingness by $\rho = \max_i \left(\sum_{j>i} r_j\right)/r_i$, we show that in the additive relaxation of the model the threshold is exactly $\rho = 1$, whereas the geometric four-ring counterexample family has infimum exactly the Tribonacci constant $T \approx 1.83929$ (the real root of $x^3 = x^2 + x + 1$), with the Descartes "pocket" circle – whose size threshold involves the golden ratio – as the obstruction. We conjecture that the geometric threshold equals T. We complement these results with a phase diagram for the divergence of the two objectives and a decoupling of the problem's hardness into a geometric layer (circle packing) and a combinatorial layer (subset sum), of which superincreasingness eliminates exactly the latter.

1 Introduction

Drop a batch of squid rings into a frying pan and a natural optimization problem appears: place as many rings as possible flat on the pan – or, if one cares about searing, maximize the total ring–pan contact area – exploiting the fact that a sufficiently small ring fits inside the hole of a larger one. Abstracted, this is the problem of packing annuli of common width into a disk with recursive nesting, a circular-container relative of the *Recursive Circle Packing Problem* (RCPP) introduced by Pedroso, Cunha and Tavares [1] to model the "telescoping" of tubes in shipping

*Draft. Computational verification scripts for every numerical claim are available in the accompanying repository.

containers, and attacked exactly by Gleixner, Maher, Müller and Pedroso [2]. That literature is algorithmic: it packs a fixed set of rings into as few rectangular containers as possible. Here we ask structural questions instead: which objectives does the natural greedy algorithm optimize, when is the choice of *where* to put each ring irrelevant, and what conditions on the radii govern the answers.

Our starting point is a classical phenomenon in one dimension. For bin packing with *divisible* item sizes, Coffman, Garey and Johnson [3] proved that First Fit Decreasing is optimal. No analogue was known for geometric packing of disks, let alone with nesting. We prove one, with *superincreasing* radii (each radius exceeding the sum of all smaller ones) playing the role of divisibility, and we determine exactly how far the hypothesis can be relaxed – with a surprising answer: the additive relaxation of the model has threshold exactly 1, while the geometric model resists up to (conjecturally, exactly) the Tribonacci constant.

Contributions. (1) A *superadditivity dichotomy*: under superincreasing radii the selection greedy is optimal for every monotone superadditive value function of the radius – contact area is one – while cardinality (not superadditive) admits counterexamples even under superincreasing radii (Section 3). (2) A *placement-obliviousness theorem*: under superincreasing radii, *any* rule for choosing among feasible containers yields the lexicographically maximal feasible set; the proof is container-shape- and dimension-agnostic (Section 4). (3) *Sharpness*: placement obliviousness holds unconditionally for $n \le 3$ rings and fails at $n = 4$; twin instances with an identical prefix rule out every state-based placement rule, deterministic or randomized (Section 5). (4) *Thresholds*: the additive model has threshold exactly $\rho = 1$; the geometric four-ring family has infimum exactly the Tribonacci constant T, and we conjecture the geometric threshold equals T (Section 6). (5) A phase diagram for the divergence of the two objectives, and a decoupling of hardness into a geometric and a combinatorial layer (Sections 7–8).

2 Model and preliminaries

Definition 1 (Rings and placements)**.** Fix a width $w > 0$. A *ring* of outer radius $r > 0$ is an annulus with inner (hole) radius $\max(0, r - w)$; if $r \le w$ it is a solid disk with no hole. Let $K \subset \mathbb{R}^d$ be a compact *container* (the *pan*; our motivating case is a disk of radius R in the plane, $d = 2$, but Theorems 6 and 8 hold for arbitrary K and d, with rings read as spherical shells). A *placement* of a subset S of the rings is a forest: each placed ring has as parent either the pan or a placed ring with $r_{\text{child}} \le r_{\text{parent}} - w$ (it sits inside the parent's hole); the children of a common parent must be packable as pairwise-disjoint balls (by outer radius) inside the parent's region — K itself, or the hole ball of radius $r - w$. Feasibility is a property of the assignment (siblings may be rearranged freely inside their container).

Definition 2 (Objectives; superincreasing radii)**.** For radii $r_1 > r_2 > \cdots > r_n$, the *contact area* of a placed ring is $a(r) = \pi\big(r^2 - \max(0, r - w)^2\big)$; the two objectives are $N = |S|$ and $A = \sum_{i \in S} a(r_i)$ over feasible S. The radii are *superincreasing* if $r_i > \sum_{j>i} r_j$ for all i; the violation is measured by $\rho = \max_i \big(\sum_{j>i} r_j\big)/r_i$. The *lex-max* set L is defined greedily: scanning $i = 1, \ldots, n$, include i whenever $(L \cap \{1, \ldots, i-1\}) \cup \{i\}$ is feasible. Since feasibility is downward closed, L is the unique lexicographically maximal feasible set, and the *selection greedy* (add ring i if the resulting *set* remains feasible) computes it. A *descending greedy with placement rule* processes rings in decreasing radius and places each into *some* feasible container (the pan or the hole of an already placed ring), skipping it only if no container admits it.

Lemma 3 (Row lemma)**.** *Balls with radii $\rho_1, \ldots, \rho_k$ satisfying $\sum_j \rho_j \le C$ pack inside a ball of radius C: place them tangent in a row along a diameter. Consequently, if a ball of radius m is removed from any packing, any collection of balls of total radius at most m can be inserted into the vacated ball without disturbing the rest.*

Proof. With centers on a diameter at coordinates $x_j = -C + 2\sum_{l<j} \rho_l + \rho_j$, each satisfies $|x_j| \le C - \rho_j$. □

3 The superadditivity dichotomy

Lemma 4. *The contact area a is increasing and superadditive on $(0, \infty)$: $a(x)+a(y) \le a(x+y)$.*

Proof. Monotonicity is clear. For $x, y \ge w$: $\pi w(2x - w) + \pi w(2y - w) = \pi w\big(2(x+y) - 2w\big) \le \pi w\big(2(x+y) - w\big)$. For $x, y \le w$ with $x + y \le w$: $x^2 + y^2 \le (x + y)^2$. For $x, y \le w < x + y$: $x^2 + y^2 \le w(x + y) \le 2w(x + y) - w^2$, using $x^2 \le wx$, $y^2 \le wy$ and $x + y \ge w$. For $x \le w \le y$: $\pi x^2 + \pi w(2y - w) \le \pi w(2x + 2y - w)$ since $x^2 \le 2wx$. □

Lemma 5 (Lexicographic dominance)**.** *Let the radii be superincreasing and let v be any positive, increasing, superadditive function of the radius. If S, T are feasible sets and the first index (in decreasing radius order) at which they differ belongs to S, then $\sum_S v(r_i) \ge \sum_T v(r_i)$, with equality only if $T \subseteq S$.*

Proof. Let i be the first differing index and $P = S \cap \{1, \ldots, i - 1\} = T \cap \{1, \ldots, i - 1\}$. If $T \setminus P = \emptyset$ then $T \subseteq S$ and monotonicity of the sum under inclusion suffices. Otherwise $T \setminus P \subseteq \{i+1, \ldots, n\}$, so $\sum_{T\setminus P} r_j < r_i$ by superincreasingness, and by iterated superadditivity and monotonicity $\sum_{T\setminus P} v(r_j) \le v\big(\sum_{T\setminus P} r_j\big) < v(r_i)$. Hence $\sum_T v = \sum_P v + \sum_{T\setminus P} v < \sum_P v + v(r_i) \le \sum_S v$. □

Theorem 6 (Selection greedy; arbitrary container, any dimension)**.** *With superincreasing radii, the selection greedy computes the lex-max feasible set and therefore maximizes $\sum_{i\in S} v(r_i)$ for every positive increasing superadditive v; in particular it maximizes the contact area A.*

Proof. Feasibility is downward closed, so the selection greedy computes L (if it could add $i \notin L$, any maximal feasible extension would lexicographically beat L). Lemma 5 shows L dominates every feasible set. □

Proposition 7 (Cardinality is not rescued)**.** *There are superincreasing instances on which every descending greedy is suboptimal for N. Example (verified): pan $R = 10$, $w = 4.8$, radii* $\{9.95, 5.0, 4.3, 0.6\}$*. Greedy places* $\{9.95, 5.0\}$ *($N = 2$, optimal area), while* $\{5.0, 4.3, 0.6\}$ *packs in a row in the pan ($N = 3$). The obstruction is structural: N corresponds to $v \equiv 1$, which is not superadditive, and Lemma 5 fails for it.*

4 Placement obliviousness under superincreasing radii

Theorem 8 (Placement rule irrelevance)**.** *Let the radii be superincreasing (weakly: $r_i \ge \sum_{j>i} r_j$ suffices) and let $K \subset \mathbb{R}^d$ be an arbitrary container. Every descending greedy, with an* arbitrary *rule for choosing among feasible containers, places exactly the lex-max feasible set. In particular best fit, worst fit and random placement are all optimal for every monotone superadditive objective.*

Proof. Let F denote the greedy's placement just before processing ring i, with placed set G, and suppose $G \cup \{i\}$ is feasible via a witness placement P; it suffices to show some container of F admits i (the converse direction is immediate, and then greedy's set equals L by induction as in Theorem 6). The available containers – the pan plus one hole per ring of G – coincide in F and P, and in P the smallest ring i parents nobody.

Let m be the largest ring of G assigned different containers by F and P (if none, P restricted to G equals F and i's container in P witnesses admission). Let u be F's container for m and v be P's. Since the greedy processes rings in decreasing order, when F placed m the occupants of u were exactly the rings larger than m that F assigns to u; by maximality of m these coincide with the rings larger than m that P assigns to u, and F certified that this set together with m packs in u. Build P' from P: (i) every ring smaller than m that P kept in u is moved into the ball of radius r_m vacated by m inside v – they fit in a row by Lemma 3, because their total radius is at most $\sum_{j:\, r_j<r_m} r_j \le r_m$, and they touch nothing else in v; (ii) m moves to u, joining exactly the larger-than-m rings whose compatibility F certified. Nesting constraints persist: u and v are the pan or holes of rings larger than m, which do not move, and every ring moved into v is smaller than r_m, which fitted in v. Thus P' is a feasible witness agreeing with F on all rings of radius $\ge r_m$: the first disagreement strictly shrinks. Iterating at most $|G|$ times yields a witness P^* agreeing with F on all of G and placing i in some container c^*; since the occupants of c^* in P^* are exactly those in F, the greedy admits i there. □

Remark 9. The proof uses only (a) downward closure of packings, (b) the row lemma applied *inside the vacated ball*, and (c) the total-radius bound from superincreasingness. Nothing refers to the shape of K or the dimension: the theorem covers rectangular pans (griddles) and spherical shells nested in arbitrary containers in $\mathbb{R}^d$ verbatim. Computational corroboration of the strongest consequence: on 100 random superincreasing instances, best-fit, worst-fit and random placement produced identical, optimal outcomes without exception.

5 Sharpness: the $n = 4$ transition and twin instances

Proposition 10 ($n \le 3$, disk pan)**.** *For a disk pan and arbitrary radii, every descending greedy on at most three rings places the lex-max set.*

Proof. The only branching point is ring r_2, when both the pan ($r_1 + r_2 \le R$; two circles fit in a disk iff their radii sum to at most its radius) and the hole of r_1 ($r_2 \le r_1 - w$) admit it. Let G be the greedy's set after r_2 and suppose $G \cup \{r_3\}$ is feasible via a witness P; we check that the greedy's placement F admits r_3 in each of the four combinations of choices. If F nested r_2: should P nest it too, the containers of F and P have identical occupants and P's home for r_3 works in F; should P keep r_2 in the pan, then $r_1 + r_2 \le R$, so $r_1 + r_3 < R$ and F's pan (holding only r_1) admits r_3. If F kept r_2 in the pan: should P do likewise, the placements agree; should P nest r_2, then $r_2 \le r_1 - w$, hence $r_3 < r_2 \le r_1 - w$ and F's *empty* hole of r_1 admits r_3. In every case r_3 is admitted, and rings outside the lex-max set are never admissible, so the greedy's set is the lex-max set. □

Lemma 11 (Cap confinement)**.** *Circles of radii* 10, 4.9, 4.8 *do not pack in a disk of radius* 15.

Proof. Write c_A, c_X, c_Y for the centers, measured from the pan center. Then $|c_A| \le 5$, $|c_X| \le 10.1$, $|c_Y| \le 10.2$, $|c_A - c_X| \ge 14.9$, $|c_A - c_Y| \ge 14.8$, $|c_X - c_Y| \ge 9.7$. From $|c_A| + |c_X| \ge 14.9$, $a := |c_A| \in [4.8, 5]$. Let $u = -c_A/|c_A|$. Then $c_X \cdot u = \big(|c_X - c_A|^2 - |c_X|^2 - a^2\big)/(2a) \ge (120 - a^2)/(2a) \ge$

9.5 (decreasing in a), so the transverse part of c_X has norm at most $\sqrt{10.1^2 - 9.5^2} \leq 3.43$; similarly $c_Y \cdot u \geq (115 - a^2)/(2a) \geq 9.0$ with transverse norm at most $\sqrt{10.2^2 - 9^2} = 4.8$. Hence $|c_X - c_Y|^2 \leq 10.1^2 + 10.2^2 - 2(9.5 \cdot 9.0 - 3.43 \cdot 4.8) \leq 67.98$, i.e. $|c_X - c_Y| \leq 8.25 < 9.7$, a contradiction. $\square$

Theorem 12 (Failure at $n = 4$). *Placement obliviousness fails for four rings: with $R = 15$, $w = 0.3$ and radii $\{10, 5, 4.9, 4.8\}$ (all four form the lex-max set, witnessed by the pan pair $\{10, 5\}$ at exact tangency and the hole pair $\{4.9, 4.8\}$ filling the hole 9.7 exactly), best fit nests the 5, forces the 4.9 into the pan, and blocks the 4.8 everywhere by Lemma 11 and the exact two-circle conditions; worst fit places all four.*

Theorem 13 (Twin instances; no state-based rule). *Let $R = 15$, $r_1 = 10$, $r_2 = 5$, $w = 0.505$ (hole 9.495), and*

$$I_1 = \{10,\ 5,\ 4.99,\ 4.50\}, \qquad I_2 = \{10,\ 5,\ 4.76,\ 4.74\}.$$

In I_1 the correct placement of the 5 is the pan (best fit fails, worst fit succeeds); in I_2 it is the hole (worst fit fails, best fit succeeds). At the decisive step the observable state – containers, capacities, occupants, incoming ring, R and w – is identical in I_1 and I_2. Consequently no deterministic placement rule that is a function of the state attains the lex-max set on all instances, and for every randomized rule some instance is failed with probability at least 1/2.

Proof ingredients. Feasibilities are two-circle-exact except for two triples, both settled rigorously: $\{10, 4.99, 4.50\}$ does not pack in the disk of radius 15 (cap confinement as in Lemma 11 yields $|c_X - c_Y| \leq 8.80 < 9.49$), and the pan pair $\{10, 5\}$ is diametrically rigid, whence no third circle of radius ≥ 4.7 fits (confinement again: for $z = 4.76$, $|c_z - c_B| \leq \sqrt{204.86 - 20 \cdot 8.8} \leq 5.37 < 9.76$). Packability of $\{10, 4.76, 4.74\}$ in radius 15 is exhibited by an explicit boundary configuration (angles 180°, 30.8°, −32.0°; all pairwise constraints verified). The key phenomenon enabling the twins is that triple-packability is *not monotone in the sum*: the unbalanced pair $\{4.99, 4.50\}$ (sum 9.49) fails beside the 10 while the balanced pair $\{4.76, 4.74\}$ (sum 9.50) succeeds. $\square$

Remark 14. The obstruction is informational, not existential: following any witness of the lex-max set is itself a legal greedy execution (a ring outside the lex-max set is never admissible), so some execution always succeeds; what cannot exist is a state-based rule that finds it. For rules with access to the full input the question becomes one of computational complexity and remains open (Section 10).

6 Thresholds: additive sharpness and the Tribonacci floor

Theorem 15 (Additive model: threshold exactly 1). *Replace sibling feasibility by its additive surrogate (a set of siblings fits iff their radii sum to at most the container capacity; exact for at most two circles and sufficient in general by Lemma 3). Then placement obliviousness holds iff $\rho \leq 1$: the proof of Theorem 8 applies verbatim when $\rho \leq 1$, and for every $\varepsilon > 0$ there is a failing instance with $\rho \leq 1 + \varepsilon$. Indeed, with r_1 fixed, $s = R - r_1 \in \left(\frac{2}{3}(r_1 - w),\ r_1 - w\right]$, $w > s/2$, take $r_2 = s$ and a pair $r_3 \approx r_4 \approx s/2$ with $r_3 + r_4 = s + \varepsilon$: the witness fills the pan with $\{r_1, r_2\}$ and the hole with the pair, while best fit nests r_2, sends r_3 to the pan, and blocks r_4 additively ($r_1 + r_3 + r_4 > R$; $r_2 + r_4 > r_1 - w$; $r_4 > r_2 - w$). All ratios tend to 1; verified instances reach $\rho = 1.03$ and 1.02.*

Proposition 16 (Geometric four-ring family: Tribonacci floor)**.** *Normalize* $r_1 = 1$*,* $t = r_2$*, and consider the* $n = 4$ *counterexample family of Theorem 12 in the tangent limit* $R = 1 + t$*,* $r_3 \to t$*,* $w \to 0$*. Its two defining pressures are: non-packability of the pan triple requires* r_4 *to exceed the Descartes pocket of the tangent pair, giving* $\rho_2 := (r_3 + r_4)/r_2 \ \geq \ 1 + \frac{1+t}{1+t+t^2}$*, while the witness pair must fit the hole, giving* $\rho_2 \leq 1/t$*. These are compatible iff* $t^3 + t^2 + t \leq 1$*, and the family infimum of* ρ *is attained where the bounds meet:*

$$\inf \rho \ = \ \frac{1}{t^*} \ = \ T, \qquad t^{*3} + t^{*2} + t^* = 1, \qquad T^3 = T^2 + T + 1,$$

the Tribonacci constant $T \approx 1.83929$*. (The pocket radius of circles* $A \geq B$ *at* $R = A{+}B$ *is exactly* $AB(A{+}B)/(A^2{+}AB{+}B^2)$ *by the Descartes circle theorem, and satisfies the curious dichotomy pocket* $\geq A/2 \iff A \leq \varphi B$ *with* φ *the golden ratio, which quantifies why comparable large rings always rescue small ones.) Margin-constrained numerical optimization over the family corroborates the floor (best verified failing instance with strict margins:* $\rho = 1.906$*; best absolute concrete failures: the twins,* $\rho = 1.898$*).*

Conjecture 17 (Tribonacci threshold)**.** *In the geometric model, placement obliviousness holds for all instances with* $\rho < T$ *and fails for suitable instances at every* $\rho > T$*; that is, the geometric threshold is exactly the Tribonacci constant, and* $T - 1$ *measures the extra protection that disk geometry (Descartes pockets) grants the greedy over additive bookkeeping. Evidence: the additive* $\rho \to 1$ *family is rescued geometrically (its small rings are half the pocket scale); structured searches restricted to* $\rho < 1.8$ *(*120 *five-ring instances, best and worst fit versus the lex-max oracle) and more than* 900 *further runs produced no failure below* T*; and a mechanism analysis: pair certificates are additively exact (hence pocket-rescued), triple certificates yield exactly* T*, quadruple certificates require pocket-scale rings that cannot share any hole as a triple, and deep nesting reproduces the same scale-invariant algebra with the width only raising the floor.*

7 The two objectives diverge: a phase diagram

Contact area behaves like $2\pi w \sum r_i$ minus a per-ring penalty πw^2; cardinality counts rings. With uniform width the hole of a large ring is large, so nesting usually reconciles the two objectives, and the divergence survives only marginally: small rings that fit k times in the pan but $k - 1$ times in the large ring's hole. The minimal instance is $R = 10$, $w = 1$, radii $\{9.0, 4.2, 4.2, 4.2\}$: the area optimum is the 9.0 with one nested 4.2 ($N = 2$, $A \approx 76.7$), the cardinality optimum the three 4.2's ($N = 3$, $A \approx 69.7$). Figure 1 maps the divergence region for the family "one large ring b plus unlimited equal small rings ρ": a staircase governed by the proved optimal thresholds for n equal circles in a disk [7, 8, 9, 10, 11]; its upper edge is exactly the three-circle threshold $0.4641R$.

8 Hardness decouples into two layers

If all radii lie within a band of width less than w, no nesting is possible and the decision problem "do all rings fit?" is exactly circle packing of the outer radii into a disk; deciding whether given circles fit into a square container is NP-hard [4], membership in NP being open because tight packings may require coordinates without short descriptions, and the disk-container version inherits the same status in the packing literature. Independently, even the *additive surrogate* is weakly NP-hard: with a no-nesting radius band and pan capacity R, maximizing $A \approx 2\pi w \sum r_i$ subject to $\sum r_i \leq R$ is subset sum. Superincreasing radii eliminate exactly this combinatorial layer – precisely as in the superincreasing knapsack [14] – leaving only the geometric one: by Theorem 6, the area optimum then reduces to n calls to a sibling-packing feasibility oracle. Useful blackboxes for the remaining geometry include the critical-density theorem for disks in a disk (any family of total area at most half the container's packs, and 1/2 is tight) [5] and Split Packing [6].

9 Generalizations

The model invites several generalizations, all of which we record here both as context for the theorems (two of them hold with no extra work) and as a research program.

(a) Container shape and dimension. Theorems 6 and 8 are stated and proved for arbitrary containers in $\mathbb{R}^d$: rectangular griddles, and tubes/spherical shells in three dimensions – the original RCPP setting – are covered verbatim, because the exchange argument acts inside the vacated ball of the moved ring only. The sharpness results (Section 5) are disk-specific; their analogues for squares and in $\mathbb{R}^3$ are open.

(b) Variable width. With per-ring widths w_i, a large ring may have a small hole, which destroys the "nesting rescue" and should widen the divergence band of Section 7 dramatically; the superadditivity lemma must be re-examined since a depends on both r and w.

(c) Flexible rings. Real squid rings bend: a ring partially resting on another still touches the pan outside the overlap, up to a lifted ramp of width δ around each overlap. The idealization $\delta = 0$ (contact area equals the annulus minus the overlapped region) defines a continuous relaxation in which partial placements trade contact for cardinality; $\delta > 0$ penalizes overlaps by a dead band proportional to the overlap perimeter.

(d) Infinite inventories. Unlimited supplies (already used in the phase diagram) suggest degenerate limits: maximal nested chains r, $r-w$, $r-2w, \dots$, density questions, and Apollonian-type limit configurations.

(e) Objectives. Theorem 6 covers every monotone superadditive value; weighted counts and concave values fall outside and, by Proposition 7, genuinely so.

10 Open problems

Open Problem 18. Prove Conjecture 17, plausibly via a universal reinsertion lemma for $\rho < T$ built from the Descartes-pocket rescue, and give a fully rigorous (non-idealized) proof of the family floor in Proposition 16.

Open Problem 19. Can a placement rule with access to the full input attain the lex-max set with polynomially many calls to a sibling-packing oracle, or is there a complexity-theoretic obstruction?

Open Problem 20. Characterize the divergence band of Section 7 exactly; determine the approximability of the cardinality objective; and decide whether the additive surrogate admits a pseudopolynomial algorithm for integer radii.

A Verification details for Theorem 13

All feasibilities in the twin instances are two-circle-exact except three facts, proved here; every numerical claim is also reproduced by the scripts in the repository (`code/gemelas.py`).

(V1) $\{10, 4.99, 4.50\}$ **does not pack in a disk of radius 15.** As in Lemma 11: $|c_A| \leq 5$, $|c_X| \leq 10.01$, $|c_Y| \leq 10.5$, $|c_A - c_X| \geq 14.99$, $|c_A - c_Y| \geq 14.5$, $|c_X - c_Y| \geq 9.49$; the first two give $a := |c_A| \geq 4.98$. With $u = -c_A/|c_A|$, $c_X \cdot u \geq (124.5 - a^2)/(2a) \geq 9.95$ and $c_Y \cdot u \geq (100 - a^2)/(2a) \geq 7.5$ (both decreasing in a, evaluated at $a = 5$), with transverse norms at most $\sqrt{10.01^2 - 9.95^2} \leq 1.095$ and $\sqrt{10.5^2 - 7.5^2} \leq 7.349$. Hence $|c_X - c_Y|^2 \leq 10.01^2 + 10.5^2 - 2(9.95 \cdot 7.5 - 1.095 \cdot 7.349) \leq 77.30$, so $|c_X - c_Y| \leq 8.80 < 9.49$.

(V2) With the pan pair $\{10, 5\}$ **placed, no third circle of radius** $z \in \{4.74, 4.76\}$ **fits.** From $|c_A| \leq 5$, $|c_B| \leq 10$ and $|c_A - c_B| \geq 15$ the triangle inequality forces equalities: the pair is diametrically rigid, $c_B = 10u$ for a unit vector u and $c_A = -5u$. For a third circle of radius z: $c_z \cdot u \geq \big((10+z)^2 - (15-z)^2 - 25\big)/10$, which equals 8.8 for $z = 4.76$ and 8.7 for $z = 4.74$; then $|c_z - c_B|^2 = |c_z|^2 - 20\, c_z \cdot u + 100 \leq (15 - z)^2 - 20\, c_z \cdot u + 100$, giving $|c_z - c_B| \leq 5.38 < 9.76$ and $\leq 5.60 < 9.74$ respectively.

(V3) $\{10, 4.76, 4.74\}$ **packs in a disk of radius 15.** Explicit boundary configuration: $c_A = (-5, 0)$, $c_X = 10.24(\cos 30.77°, \sin 30.77°) \approx (8.799, 5.238)$, $c_Y = 10.26(\cos 32.02°, -\sin 32.02°) \approx (8.700, -5.440)$. Then $|c_A| = 5$, $|c_X| = 10.24 = 15 - 4.76$, $|c_Y| = 10.26 = 15 - 4.74$ (all inside); $|c_A - c_X| = 14.76$ and $|c_A - c_Y| = 14.74$ (exact tangencies, chosen via the boundary angles); and $|c_X - c_Y| \approx 10.68 \geq 9.50$ with ample margin.

(V4) Hole facts. The hole of the 10 has radius 9.495; $4.99 + 4.50 = 9.49 \leq 9.495$ (the pair fits, two-circle exact) while $4.76 + 4.74 = 9.50 > 9.495$ and $5 + 4.50 = 9.50 > 9.495$ (blocked, two-circle exact).

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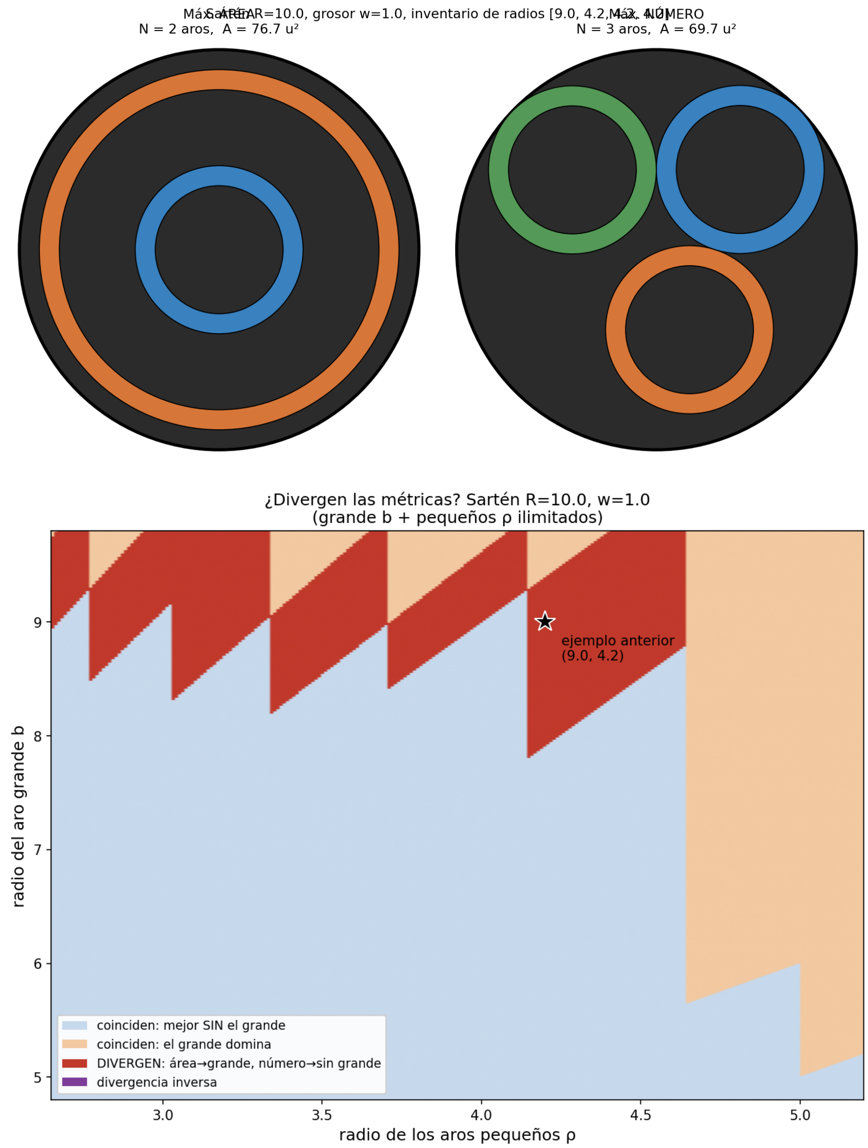

Figure 1: Top: the minimal divergence instance; area optimum (left) versus cardinality optimum (right). Bottom: phase diagram of the divergence band for uniform width.

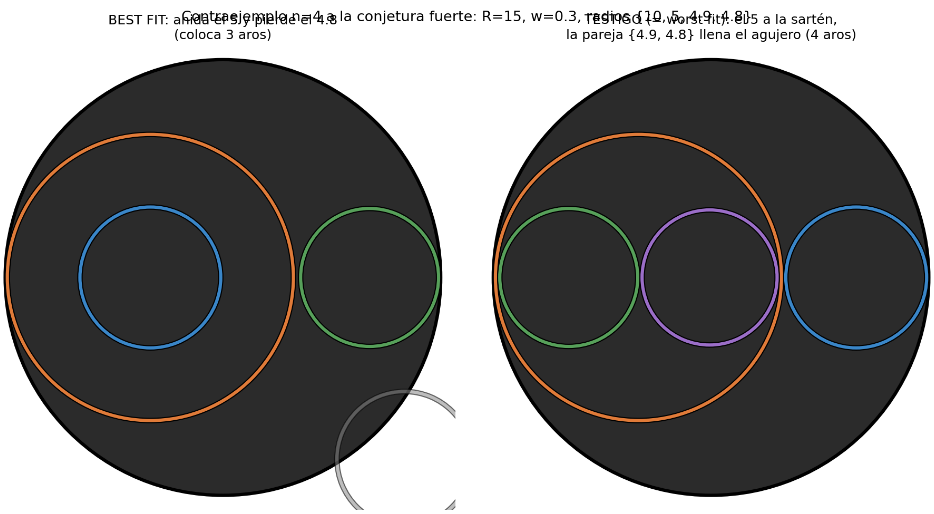

Figure 2: The $n = 4$ counterexample of Theorem 12: best fit (left) nests the 5 and loses the 4.8; the witness (right), realized by worst fit, places all four rings.